

Electromagnetic Fields

(3) Electric Flux and Gauss Law

2021/2022

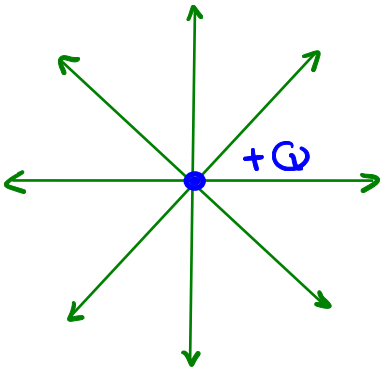
Chapter (2)

The electric flux density and Gauss's law

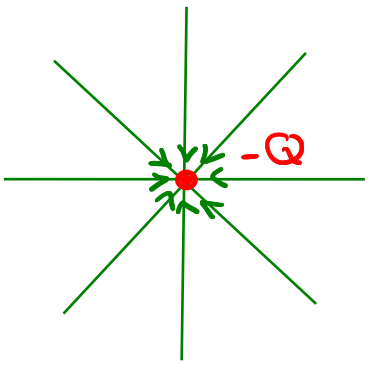
-Electric flux: the number of electric field lines that intersect a given area

-Properties:

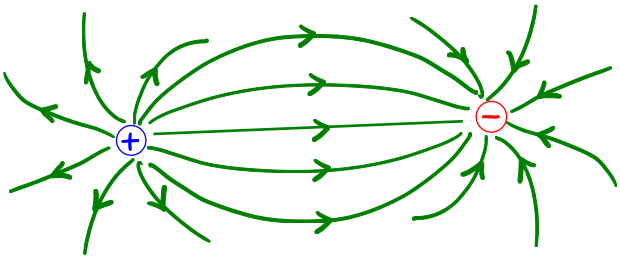
- the electric flux generates from the positive charge $+Q$ (C)



- Electric flux is directed toward the negative charge $-Q$ (C)



- Electric flux come out from positive charge and terminates on negative charge



- Electric flux is in the same direction as Electric field intensity "E"

- In the SI system of units, the total flux released from a charge of Q(C) equals Q(C)

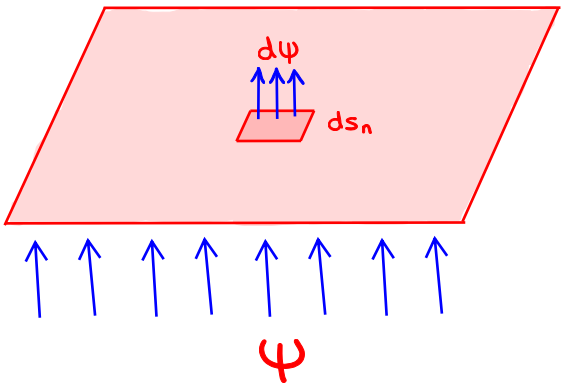
- A single flux line is released from 1C of charge

$$\Psi = Q \ C$$

• Electric flux density ($\vec{D}$) :

⇒ is defined as the number of electric flux lines per unit normal area

- $|\vec{D}| = \frac{d\Psi}{ds}$ (lines/m² or C/m²)
- $\Psi = \iint_s \vec{D} \cdot \vec{ds}$



Electric flux density due to conducting metal sphere :

- the electric flux density is in radial direction

∴ $\vec{D}|_a = \frac{Q}{4\pi a^2} \hat{a}_r$ (Just outside inner Sphere)

∴ $\vec{D}|_b = \frac{Q}{4\pi b^2} \hat{a}_r$ (Just Before outer Sphere)

"كسان تفهم ازاى اقرأ الخطوات اللى فى الجنب"

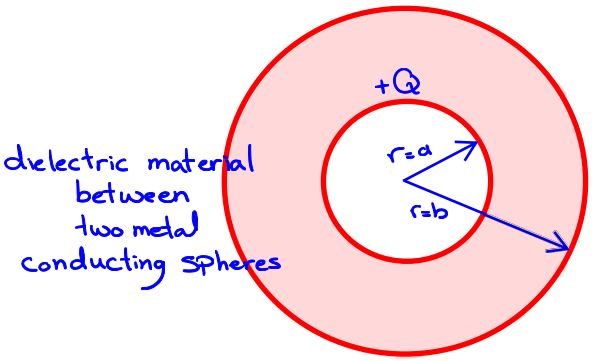
⇒ In general : $\vec{D} = \frac{Q}{4\pi r^2} \hat{a}_r \ C/m^2$

∴ $\vec{E} = \frac{Q}{4\pi \varepsilon_o r^2} \hat{a}_r$

∴ from the above equation :

$$\vec{D} = \varepsilon_o \vec{E} \quad C/m^2$$

Generally, $\vec{D} = \varepsilon_o \varepsilon_r \vec{E}$

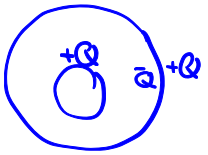


1) • Outer sphere is grounded
"$Q_{net-outersphere} = 0$"

2) Inner sphere is charged with +Q

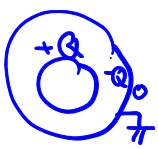
3) A charge -Q is induced on the inner-surface of the outer sphere
"It is called bounded charge that can't be discharged"

3) Now,



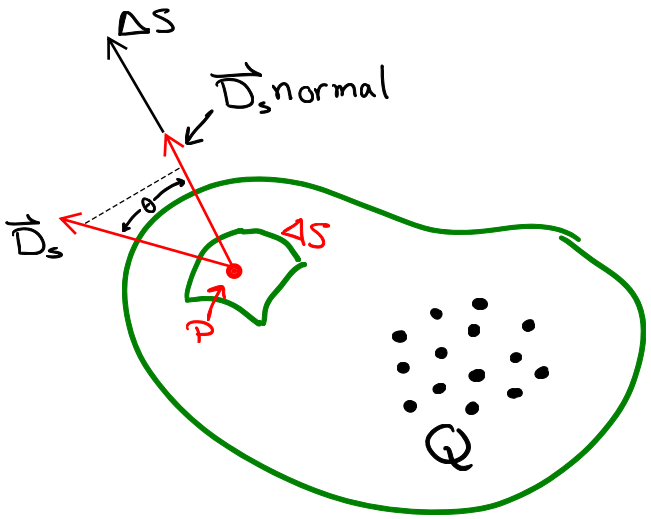
The net charge is +Q

4) If the outer sphere is now grounded



• Gauss's Law :

it states that : "the electric flux passing through any closed surface is equal to the total charge enclosed by that surface".



∴ $\Delta\psi$ = flux crossing ΔS = $D_{s\,norm}\,\Delta S = D_s \cos\theta\,\Delta S$

$$\Delta\psi = \vec{D} \cdot d\vec{s}$$

∴ $$\boxed{\psi = \oint \vec{D} \cdot d\vec{s} = Q_{enclosed}}$$

⇒ where : ds = differential surface element

$\vec{D}_s$ = Electric flux density

ψ = number of flux lines

$Q_{enclosed}$ = total charge inside Gaussian surface

-Properties : 1). the Surface is closed

2). Electric flux density $\vec{D}$ is either :

- normal to Closed Surface $\Rightarrow \vec{D}.ds = D_n ds_n$
- Tangential to Closed Surface $\Rightarrow \vec{D}.ds =$ Zero

3). $\vec{D}_{normal}$ is Constant

$\Rightarrow$ if the 3 points are satisfied

$\therefore \oint_s \vec{D}.ds = D_n \times$ Area of gaussian Surface

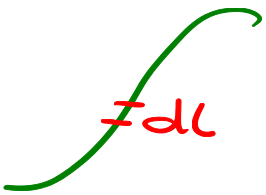
-the Charge enclosed might be :

$\Rightarrow$ Several point Charges :

- $Q = \sum Q_n$

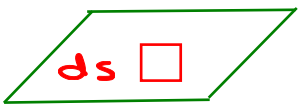
$\Rightarrow$ line Charge :

- $Q = \int \rho_L \, dL$



$\Rightarrow$ Surface Charge :

- $Q = \int_s \rho_s \, ds$



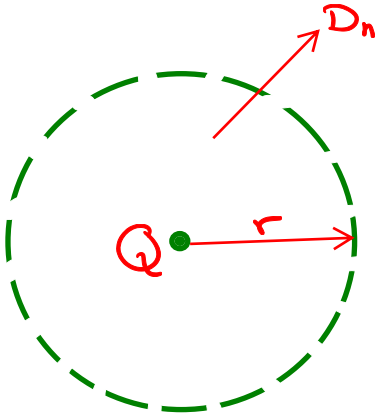
$\Rightarrow$ volume Charge distribution :

- $Q = \oint D_s.ds = \int_{Vol} \rho_v \, dV$

-Application of Gauss's law:

1). Point Charge:

- Sphere is the Gaussian surface
- Point charge at the center



Gaussian surface: sphere

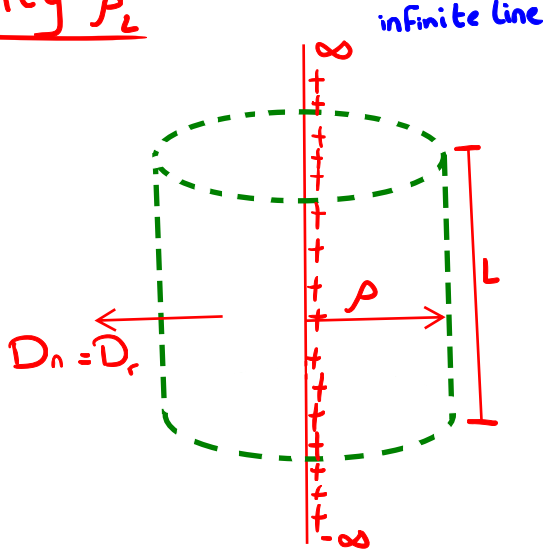
- $\oint_S \vec{D} \cdot d\vec{s} = Q_{en}$
- $D \times 4\pi r^2 = Q$, $\vec{D} = \frac{Q}{4\pi r^2} \hat{a}_r$

∴ $\vec{D} = \varepsilon_0 E$, ∴ $\vec{E} = \frac{Q}{4\pi \varepsilon_0 r^2} \hat{a}_r$

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2). Line Charge: with uniform Charge density ρ_L

- Cylinder is the Gaussian Surface
- Line Charge is the axis of Cylinder
- ρ_L must be constant
- $Q_{en} = \rho_L L$
- Area of Gaussian Surface $= 2\pi\rho L$
- $\oint_S \vec{D} \cdot d\vec{s} = Q_{en}$



Gaussian Surface: Cylinder

$\vec{D}$ موازية للسطح الفوق و التحت عشان كدة هيّ هتتعلم تكامل Dot Product = Zero

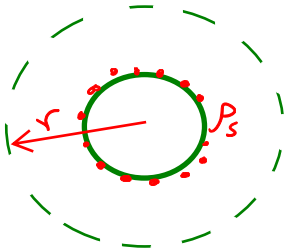
- $\iint \vec{D} \cdot d\vec{s} = \rho_L \cdot L$, $D \times 2\pi\rho L = \rho_L L$

∴ $\vec{D} = \frac{\rho_L}{2\pi\rho} \hat{a}_\rho$, ∴ $\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho} \hat{a}_\rho$

3). Infinite surface "Sheet" with uniform Charge density ρ_s

Prove it yourself

- Other Shapes for Spherical Gaussian Surface



Spherical Conductor

ρ_s = Constant

$r \le a$ & $D = 0$

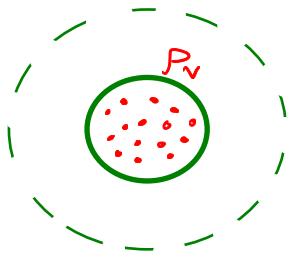
$r \ge a$ & $D(4\pi r^2) = \rho_s(4\pi a^2)$

& $\vec{D} = \begin{cases} 0 & r \le a \\ \dfrac{\rho_s a^2}{r^2} & r \ge a \end{cases}$

Hint: At surface of any Conductor:

$\vec{D} = \rho_s$

Or $\vec{E} = \dfrac{\rho_s}{\epsilon_0}$



Spherical insulator

ρ_v = Constant or f(r)

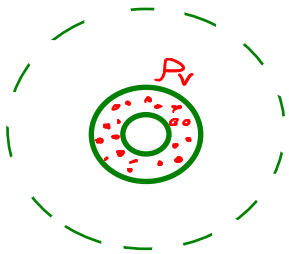
$r \le a$ & $D(4\pi r^2) = \rho_v(4\pi r^3)/3$

$r \ge a$ & $D(4\pi r^2) = \rho_v(4\pi a^3)/3$

& $\vec{D} = \begin{cases} \rho_v/3 & r \le a \\ \dfrac{\rho_v a^3}{3r^2} & r \ge a \end{cases}$

ρ_v = Const. ظبطاً داخل الحل

$Q = \int \rho_v \, dv$ غير كدا



Hollow sphere insulator

ρ_v = Constant or f(r)

$r \le a$ & $D = 0$

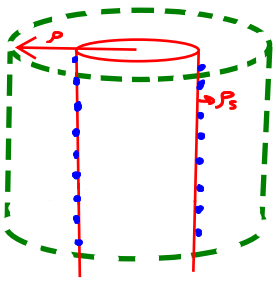
$a \le r \le b$ & $D(4\pi r^2) = \rho_v(4\pi[r^3 - a^3])/3$

$r \ge b$ & $D(4\pi r^2) = \rho_v(4\pi[b^3 - a^3])/3$

& $\vec{D} = \begin{cases} 0 & r \le a \\ \dfrac{\rho_v[r^3 - a^3]}{3r^2} & a \le r \le b \\ \dfrac{\rho_v[b^3 - a^3]}{3r^2} & r \ge b \end{cases}$

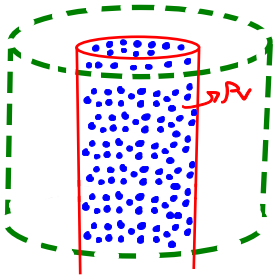
ρ_v = Const. ظبطاً داخل الحل

Other Shapes of Cylindrical Gaussian Surface:



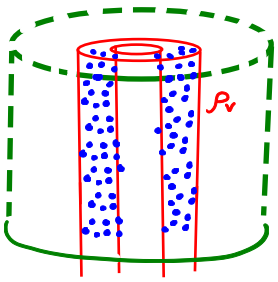
- Cylindrical Conductor

ρ_s = Constant



- Cylindrical insulator

ρ_v = Constant or f(r)



- Cylindrical Shell "insulator"

ρ_v = Constant or f(r)

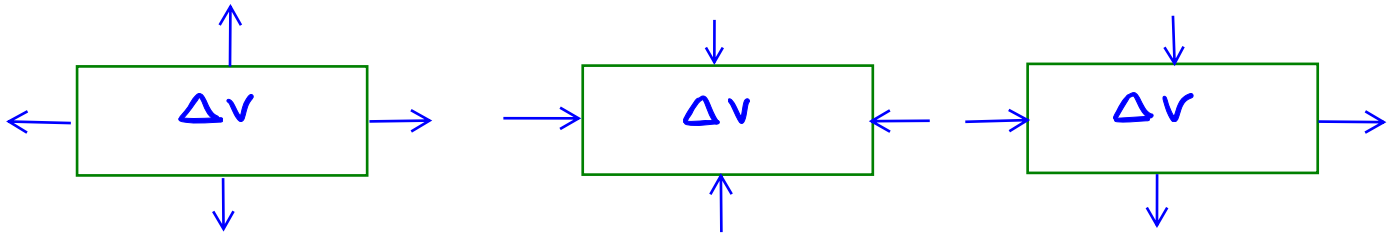
ex] $\vec{D}_{outside}$:

$D \times 2\pi r L = Q$

Conductor → $Q = \int \rho_s \, ds = \rho_s \, 2\pi a L \Rightarrow \vec{D} = \dfrac{\rho_s a}{r} \hat{a}_r$

insulator → $Q = \int \rho_v \, dv$

- Divergence: " the divergence of the vector flux $\vec{A}$ is the outflow of flux from a small closed surface per unit volume as the volume goes to zero".



+ve divergence
« source of flux »

-ve divergence
« sink of flux »

Zero divergence
« zero flux »

- The divergence tell us how flux is leaving small volume, but it doesn't describe it's direction.

- The vector operator:

⇒ del (nabla) operator $\vec{\nabla}$ is defined as vector operator

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$$

then the dot product of $\vec{\nabla}.\vec{D}$

$$\vec{\nabla}.\vec{D} = \left(\frac{\partial}{\partial x}\hat{a}_x + \frac{\partial \hat{a}_y}{\partial y} + \frac{\partial \hat{a}_z}{\partial z}\right).\left(D_x\hat{a}_x + D_y\hat{a}_y + D_z\hat{a}_z\right)$$

$$= \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

⇒ the outcome: $\text{Div}\,\vec{D} = \vec{\nabla}.\vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

is called divergence of $\vec{D}$

- For cartesian coordinates if: $\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$

c.c $\text{Div}\,\vec{A} = \bar{\nabla}\cdot\bar{A} = \dfrac{\partial A_x}{\partial x} + \dfrac{\partial D_y}{\partial y} + \dfrac{\partial D_z}{\partial z}$

- For cylindrical coordinates if: $\vec{A} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z$

$\text{Div}\,\bar{A} = \nabla\cdot\bar{A} = \dfrac{1}{\rho}\dfrac{\partial}{\partial \rho}(\rho A_\rho) + \dfrac{1}{\rho}\dfrac{\partial A_\phi}{\partial \phi} + \dfrac{\partial A_z}{\partial z}$

⇓ given

- For spherical coordinates if: $\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$

c.c $\text{Div}\,\bar{A} = \bar{\nabla}\cdot A = \dfrac{1}{r^2}\dfrac{\partial (r^2 A_r)}{\partial r} + \dfrac{1}{r\sin\theta}\dfrac{\partial (A_\theta \sin\theta)}{\partial \theta} + \dfrac{1}{r\sin\theta}\dfrac{\partial A_\phi}{\partial \phi}$

⇓ given

- Electric flux with a differential volume Element:

- The Divergence Theorem: "the integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field through out the volume enclosed by the closed surface".

$$\oint \bar{D}\cdot ds = \int_{\text{Volume}} (\bar{\nabla}\cdot\bar{D})\, dv$$

- Maxwell's first equation: it states that "the electric flux per unit volume leaving a small volume is exactly equal to the volume Charge density there".

$$(\bar{\nabla}\cdot\bar{D}) = \rho_v$$

• هيساعد في حل الاشكال غير المنتظمة عشان مش هنعرف نستخدم اشكال Gauss

- $\int D \cdot ds = Q$
- $\int D \cdot ds = \left[\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right] \Delta v$

∴ $\rho_v = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

∴ $(\bar{\nabla} \cdot \bar{D}) = \rho_v$

"Problems"

mid-2020

1). An electric flux density is Given by $\vec{D} = xy^2 \hat{a}_x - xz^2 \hat{a}_y + xyz \hat{a}_z$ C/m^2. The volume charge density at P(1,2,3) in C/m^3 is:

$\Rightarrow \rho_v = \vec{\nabla} \cdot \vec{D}$, $\vec{\nabla} \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

$\Rightarrow \vec{\nabla} \cdot \vec{D} = y^2 - 0 + xy$, at p(1,2,3)

$\rho_v = (2)^2 + (1)(2) = \boxed{6}$ C/m^3

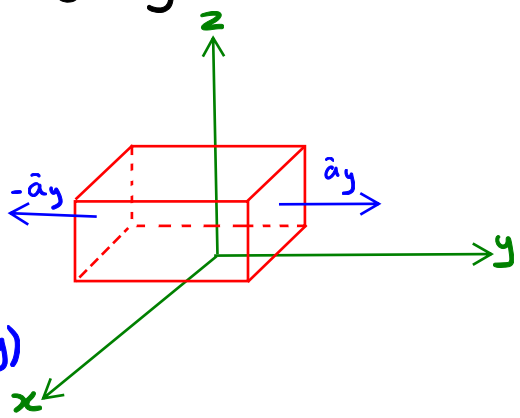
mid-2020

2). Three point charges , $Q_1 = 30nc$, $Q_2 = -150$ nc, and $Q_3 = -70$ nC are enclosed by surface S. The net flux S is?

$\Rightarrow \Psi = Q_{en} = Q_1 + Q_2 + Q_3 = 30 - 150 - 70 = -190$ C

mid-2020

3). A cube of 2 m side is centered at the origin with the edges parallel to the axes. within the cube, $\vec{D} = 40y^2\,\hat{a}_y$ C/m². the total Charge contained in the cube will be?



$$\Rightarrow Q = \int_S \vec{D}\cdot d\vec{s} = \int_{-1}^{1}\int_{-1}^{1} 40y^2(\hat{a}_y)\,dx\,dz\,(\hat{a}_y) \quad (y=1)$$

$$\oplus \int_{-1}^{1}\int_{-1}^{1} 40y^2(\hat{a}_y)\,dx\,dz\,(-\hat{a}_y) \quad (y=-1)$$

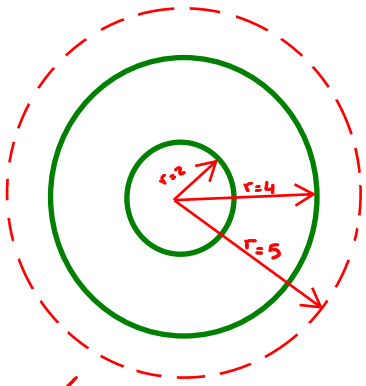
$$= 40\int_{-1}^{1}dx\int_{-1}^{1}dz - 40\int_{-1}^{1}dx\int_{-1}^{1}dz$$

$$\therefore Q_{en} = 0$$

• لو كان مركز المكعب هيبقى الـ origin الإجابة كانت هتختلف عشان المعادلة هتتغير على محور (y)

mid-2020

4). Spherical Surfaces at r=2 and 4 m Carry uniform Surface Charge densities of 20 nC/m² and -4 nC/m², respectively. the electric flux density in nC/m² at r= 5m is?



$$\Rightarrow Q_{en} = \oint D\cdot ds$$

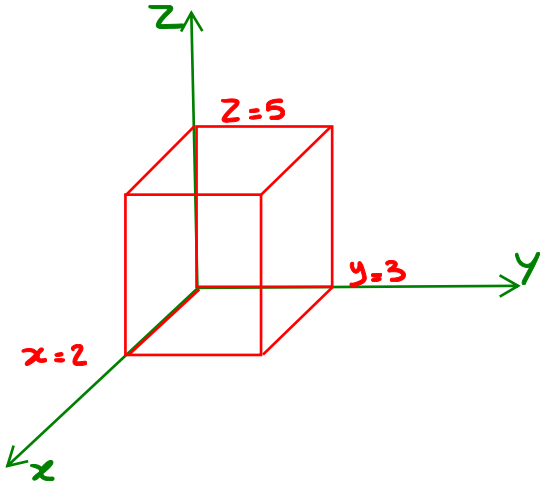
$$\Rightarrow D \times 4\pi(5)^2 = 20\times10^{-9}\times 4\pi(2)^2 - 4\times10^{-9}\times 4\pi(4)^2$$

$$\therefore \vec{D} = 0.64\,\hat{a}_r \text{ nC/m}^2$$

Problem Set #2

5). Let $\vec{D} = 4xy\,\hat{a}_x + 2(x^2+z^2)\hat{a}_y + 4yz\,\hat{a}_z$ C/m². evaluate surface integrals to find the total charge enclosed in the rectangular parallelepiped $0<x<2$, $0<y<3$, $0<z<5$ m

$$\Rightarrow \oint \vec{D}\cdot ds = \Sigma Q_{en}$$



$$\hat{a}_x : \int_0^5\int_0^3 4xy\,dy\,dz - \int_0^5\int_0^3 4xy\,dy\,dz$$

(2) ; zero ; zero

$$\hat{a}_y : \int_0^5\int_0^2 2(x^2+3^2)\,dx\,dz - \int_0^5\int_0^2 2(x^2+3^2)\,dx\,dz$$

$$\hat{a}_z : \int_0^3\int_0^2 4yz\,dx\,dy - \int_0^3\int_0^2 4yz\,dx\,dy$$

(5) ; zero ; zero

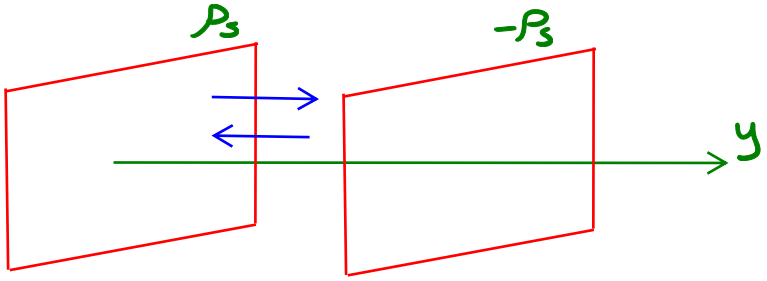
- $\hat{a}_y$ flux will cancel each other

$$\therefore \Sigma Q_{en} = \int_0^5\int_0^3 8y\,dy\,dz + \int_0^3\int_0^2 20y\,dx\,dy = \boxed{360\ C}$$

mid-2019

-6). Two sheets having surface charge density of ρ_s, $-\rho_s$ are kept parallel to each other at a small separation distance d. The electric field at any point between the sheets is.

- $\vec{E} = \frac{\rho_s}{2\varepsilon_0}(\hat{a}_y) + \frac{-\rho_s}{2\varepsilon_0}(-\hat{a}_y)$

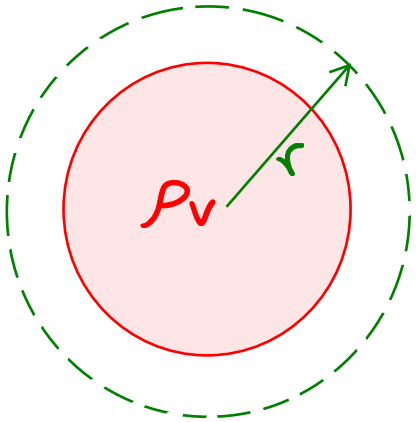


- $|\vec{E}| = \frac{\rho_s}{\varepsilon_0}$

Problem Set #1

7). Volume charge density is located in free space as $\rho_v = 2e^{-1000r}$ nC/m³ for $0<r<1$ mm, and $\rho_v = 0$ elsewhere.

(a). Find the total charge enclosed by the spherical surface r = 1mm



$$\Rightarrow Q = \iiint \rho_v \, dv$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{1\times10^{-3}} 2e^{-1000r} * r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$= 4 \times 10^{-9}$ nC El calculator على احسبة ، تكامل لازم هش

(b). by using Gauss's law, calculate the value of Dr on the surface r = 1 mm

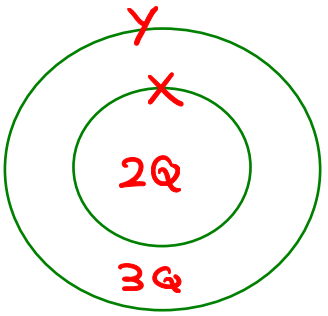
$$\Rightarrow \oint \bar{D} \cdot d\bar{s} = \Sigma Q_{en}$$

$$\Rightarrow D \times 4\pi r^2 = 4 \times 10^{-9} \quad , \quad \vec{D} = \frac{4 \times 10^{-9}}{4\pi r^2} \hat{a}_r$$

$$\Rightarrow D_r = \frac{4 \times 10^{-9}}{4\pi (1\times10^{-3})^2} = \boxed{3.2 \times 10^{-4}} \text{ nC/m}^2$$

mid-2019

8). X and Y are two hollow concentric spheres enclosing charge 2Q and 3Q respectively as shown. The ratio of electric flux through X to Y



$\Rightarrow$ From Gauss's law : $\Psi = Q_{en}$

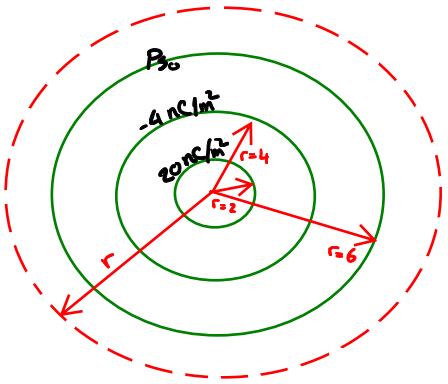
$$\Rightarrow \Psi_x = 2Q \quad , \quad \Psi_y = 2Q + 3Q = 5Q$$

$$\Rightarrow \frac{\Psi_x}{\Psi_y} = 2Q/5Q = \frac{2}{5} = \boxed{0.4}$$

Problem Set #3

9). Spherical surfaces at r = 2, 4 and 6 m carry uniform surface charge densities of 20 nC/m², -4 nC/m², and ρ_{s_0}, respectively

a). Find $\vec{D}$ at r = 1, 3, and 5 m



$$\oint \vec{D} \cdot d\vec{s} = Q_{en}$$

- r < 2, no charge is enclosed $\therefore D_r = 0$

- 2 < r < 4 : $D_r (4\pi r^2) = 4\pi (2)^2 (20 \times 10^{-9})$

$$D_r = \frac{80 \times 10^{-9}}{r^2} \ C/m^2$$

$\therefore D_r (r=3) = \boxed{8.9 \times 10^{-9}} \ C/m^2$

- 4 < r < 6 : $D_r (4\pi r^2) = 20 \times 10^{-9} \times (4\pi (2)^2) + (-4 \times 10^{-9})(4\pi (4)^2)$

$$D_r = \frac{16 \times 10^{-9}}{r^2}$$

$\therefore D_r (r=5) = 6.4 \times 10^{-10} \ C/m^2$

b). Determine ρ_{s_0} such that D = 0 at r = 7 m

$\Rightarrow D_r (r=7) = \frac{16 \times 10^{-9}}{r^2} + \frac{\rho_{s_0} (6)^2}{r^2} = 0$ "$\Sigma Q = 0$"

$\therefore \rho_{s_0} = (-4/9) \times 10^{-9} \ C/m^2$

10). A radial electric field distribution in free space is given in spherical coordinates as:

$$\mathbf{E}_1 = \frac{r\rho_0}{3\epsilon_0}\,\mathbf{a}_r \qquad (r \le a)$$

$$\mathbf{E}_2 = \frac{(2a^3 - r^3)\rho_0}{3\epsilon_0\, r^2}\,\mathbf{a}_r \qquad (a \le r \le b)$$

$$\mathbf{E}_3 = \frac{(2a^3 - b^3)\rho_0}{3\epsilon_0\, r^2}\,\mathbf{a}_r \qquad (r \ge b)$$

where ρ_0, a, and b are constants. (a) Determine the volume charge density in the entire region ($0 \le r \le \infty$) by the appropriate use of $\nabla \cdot \mathbf{D} = \rho_v$. ($b$) In terms of given parameters, find the total charge, Q, within a sphere of radius r where $r > b$.

$\Rightarrow \rho_v = \vec{\nabla}\cdot D$

$\Rightarrow D = \varepsilon_0 E$, D is in $\hat{a}_r$ direction only

$$\text{c.c. } \rho_{v1} = \vec{\nabla}\cdot\vec{D}_1 = \frac{1}{r^2}\frac{d}{dr}(r^2 D_1) = \frac{1}{r^2}\frac{d}{dr}\left(\frac{r^3\rho_0}{3}\right) =$$

$$= \frac{3r^2}{r^2 3}\rho_0 = \rho_0 \quad (r \le a)$$

$$\text{c.c. } \rho_{v2} = \frac{1}{r^2}\frac{d}{dr}(r^2 D_2) = \frac{1}{r^2}\frac{d}{dr}\left(\frac{1}{3}(2a^3 - r^3)\rho_0\right)$$

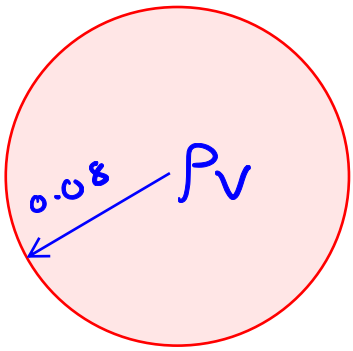
$$= \frac{-3r^2\rho_0}{r^2 3} = -\rho_0 \quad (a \le r \le b)$$

$$\text{c.c. } \rho_{v3} = \frac{1}{r^2}\frac{d}{dr}(r^2 D_3) = \frac{1}{r^2}\frac{d}{dr}\underbrace{\left(\frac{1}{3}(2a^3 - b^3)\rho_0\right)}_{\text{constants}} = 0 \quad (r \ge b)$$

b). $Q = \iiint \rho_v \, dv =$

$$= \int_0^{\pi}\int_0^{\pi}\int_0^{a} \rho_0\, r^2 \sin\theta \, dr\, d\theta\, d\phi - \int_0^{\pi}\int_0^{\pi}\int_0^{b} \rho_0\, r^2 \sin\theta \, dr\, d\theta\, d\phi = \frac{4}{3}\pi(2a^3 - b^3)\rho_0$$

11. Let $\vec{D} = 5.00 r^2 \hat{a}_r$ mC/m² for $r \le 0.08$ m and $\vec{D} = 0.205 \hat{a}_r / r^2$ μC/m² for $r \ge 0.08$ m.



(a) Find ρ_v for $r = 0.06$ m

- $\vec{D}$ is $\hat{a}_r$ direction only
- $\rho_v = \nabla \cdot \vec{D}$

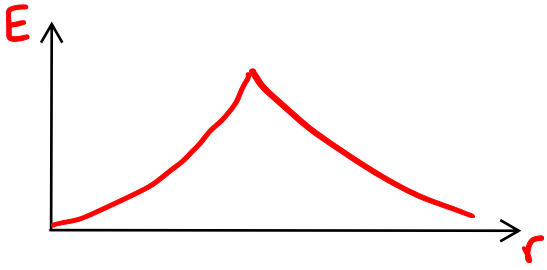
$$\rho_v = \frac{1}{r^2}\frac{\partial}{\partial r}[r^2 D_r] = \frac{1}{r^2}\frac{\partial}{\partial r}[r^2 * 5r^2] = \frac{1}{r^2}\frac{\partial}{\partial r}[5r^4]$$

$$\rho_v = \nabla \cdot \vec{D} = \frac{1}{r^2}[20r^3] = 20r \quad \text{mC/m}^3$$

∴ at $r = 0.06$ m, $\rho_v = \boxed{1.2}$ mC/m³

(b). Find ρ_v for $r = 0.1$ m

⇒ using $\vec{D} = 0.205 \hat{a}_r / r^2$



$$\Rightarrow \rho_v = \nabla \cdot \vec{D} = \frac{1}{r^2}\frac{\partial}{\partial r}[r^2 D_r] = \frac{1}{r^2}\frac{\partial}{\partial r}[0.205] = 0$$

(0.205 is a constant)

∴ $\rho_v = 20r \quad r \le 0.08$
$= 0 \quad r \ge 0.08$

(c). What surface charge density could be located at $r = 0.08$ m to cause $\vec{D} = 0$ for $r > 0.08$ m?

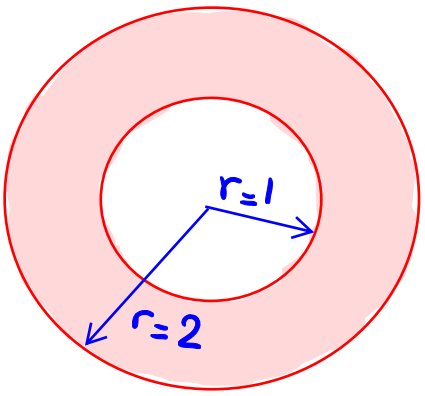
$$\Rightarrow \Sigma Q_{en} = \iiint \rho_v \, dv + \rho_s * 4\pi (0.08)^2 = 0$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{0.08} 20 r \cdot r^2 \sin\theta \, dr \, d\theta \, d\phi = 2.57 \times 10^{-3} \text{ mC} = 2.57 \ \mu\text{C}$$

∴ $\rho_s = -\frac{2.57}{4\pi(0.08)^2} = \boxed{-32 \ \mu\text{C/m}^2}$

Problem Set #6

12). For the vector field $\vec{D} = 3r^2 \hat{a}_r$, evaluate both sides of the divergence theorem for the region enclosed between the spherical shells defined by $r=1$ and $r=2$



$\Rightarrow$ to verify Divergence theorem

- $\oint \vec{D} \cdot \vec{ds} = \sum Q_{en} = \iiint \rho_v \, dv$

$\therefore \oint \vec{D} \cdot \vec{ds} = \iiint (\vec{\nabla} \cdot \vec{D}) \, dv$

- first : L.H.S : $\int_0^{2\pi} \int_0^{\pi} 3r^2 \cdot r^2 \sin\theta \, d\theta \, d\phi$

$= 12\pi (r)^4$

$= 12\pi (2^4 - 1^4)$

$= 180\pi$

- R.H.S : $\iiint (\vec{\nabla} \cdot \vec{D}) \, dv$

- $\vec{\nabla} \cdot \vec{D} = \frac{1}{r^2} \frac{\partial}{\partial r} [r^2 3r^2] = \frac{1}{r^2} [12r^3] = 12r$

$\therefore$ R.H.S $= \int_0^{2\pi} \int_0^{\pi} \int_1^2 12r \cdot r^2 \sin\theta \, dr \, d\theta \, d\phi$

$\Rightarrow$ R.H.S $= 180\pi$

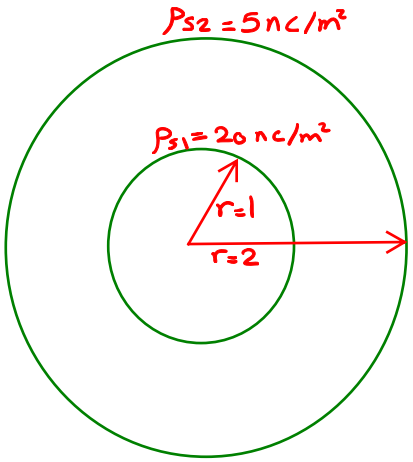
$\therefore$ L.H.S = R.H.S $= 180\pi$

mid-2018

13). Two conducting sphere shells of radii r=1 and 2 m carry a uniform charge densities of 20 nc/m² and 5 nc/m², respectively

i. Find the electric flux density in all regions

1). For r<1 ∴ Q = Zero , $D_o = 0$



2). For 1<r<2 ∴ $\oint D.ds = Q_{en}$

$\Rightarrow D \times 4\pi r^2 = \rho_{s1} \times 4\pi (1)^2$

∴ $D_1 = \frac{\rho_{s1}}{r^2} = \boxed{\frac{20 \times 10^{-9}}{r^2} \hat{a}_r}$

3). For r>2 ∴ $D(4\pi r^2) = \rho_{s1} \times (4\pi (1)^2) + \rho_{s2} (4\pi (2)^2)$

∴ $D_2 = \frac{\rho_{s1}(1)^2 + \rho_{s2}(2)^2}{r^2} = \frac{20 \times 10^{-9} + 20 \times 10^{-9}}{r^2} \hat{a}_r = \boxed{\frac{40 \times 10^{-9}}{r^2} \hat{a}_r}$

ii). Find the surface charge density located at r=3m to make D=0 when r>3 m

$\oint \vec{D}.\vec{ds} = Q_{en}$, ∴ $Q_{en} = \rho_{s1}(4\pi (1)^2) + \rho_{s2}(4\pi (2)^2) + \rho_{s3}(4\pi (3)^2) = 0$

$= 20 \times 10^{-9} + 20 \times 10^{-9} + \rho_{s3}(9) = 0$

∴ $\rho_{s3} = \boxed{\frac{-40}{9} \ nc/m^2}$

14). let $\vec{D} = \frac{r}{3}\,\hat{a}_r$ nC/m in free space.

a). Find $\vec{E}$ at $r = 0.2$ m

b). Find the total charge within the sphere $r = 0.2$ m.

c). Find the total flux leaving the sphere $r = 0.3$m

a). $\because \vec{E} = \frac{\vec{D}}{\varepsilon_0} = \frac{r}{3\varepsilon_0}\,\hat{a}_r$

$\therefore$ $\vec{E}$ at $r = 0.2$ m, $\vec{E} = \frac{0.2 \times 10^{-9}}{3 \times 8.854 \times 10^{-12}} = 7.53\,\hat{a}_r$ V/m

b). Q at $r = 0.2$, $Q = \oint_S \vec{D} \cdot d\vec{s}$

$$Q = \int_0^{2\pi}\int_0^{\pi} \frac{0.2 \times 10^{-9}}{3}\,\hat{a}_r \cdot (0.2)^2 \sin\theta\, d\theta\, d\phi\, a_r = \boxed{33.5\ \text{pC}}$$

c). Ψ at $r = 0.3$m, $\Psi = \oint \vec{D} \cdot d\vec{s}$

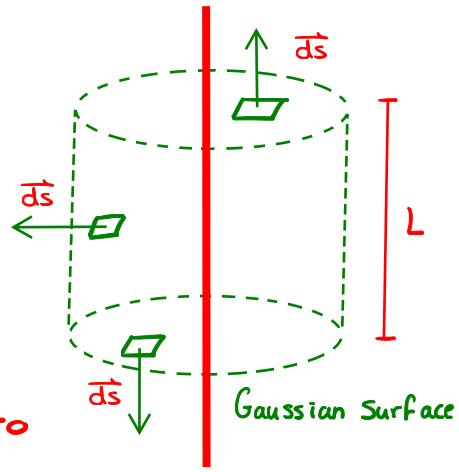
$$\Psi = \int_0^{2\pi}\int_0^{\pi} \frac{0.2 \times 10^{-9}}{3}\,\hat{a}_r \cdot (0.3)^2 \sin\theta\, d\theta\, d\phi\, a_r = \boxed{113.1\ \text{pC}}$$

mid-2019

15). The electric field intensity at a point on the surface of a conductor is given by $\vec{E} = 0.2\,\hat{a}_x + 0.3\,\hat{a}_y + 0.2\,\hat{a}_z$ V/m. Supposing the conductor to be surrounded by free space, the surface charge density at the point is?

$\Rightarrow \rho_s = D = \varepsilon_0 E = \varepsilon_0 \sqrt{0.2^2 + 0.3^2 + 0.2^2} = \boxed{3.64\ \text{pC/m}^2}$

16). Calculate $\vec{D}$ and $\vec{E}$ at a distance r due to a uniformly line charge distribution of ρ_L and infinite length



$\Rightarrow \vec{D} = D_r \hat{a}_r$

$\Rightarrow Q = \oint_S \vec{D}.ds = \oint_{sides} \vec{D}.ds + \oint_{Top} \vec{D}.ds + \oint_{bottoms} \vec{D}.ds$

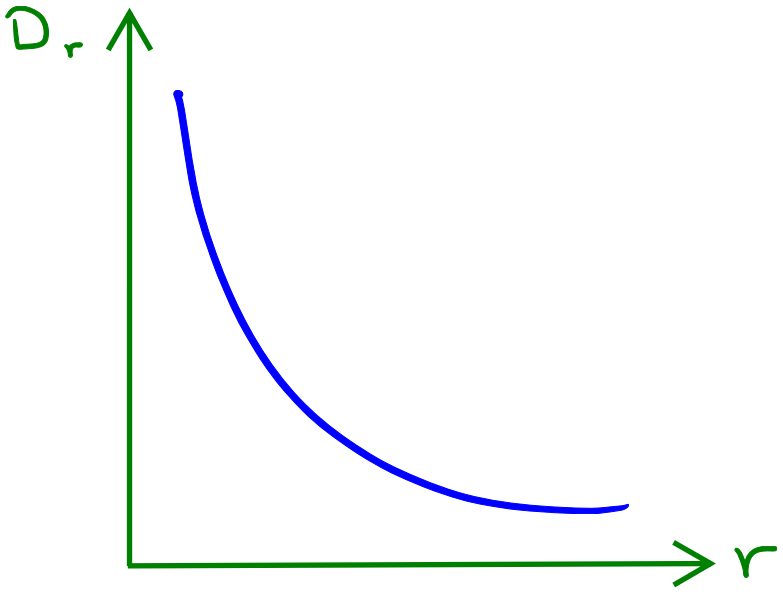
$\Rightarrow$ The integral on the top or the bottom equal zero since $\vec{D}$ and $d\vec{s}$ are normal

$\Rightarrow Q = \oint_{sides} \vec{D}.ds = D_r \int_0^L \int_0^{2\pi} r \, d\phi \, dz = D_r (2\pi r L)$

$\therefore Q = \rho_L L$

$\therefore D = \frac{\rho_L}{2\pi r} \hat{a}_r \ C/m^2$, $\vec{E} = \frac{\vec{D}}{\varepsilon_0} = \frac{\rho_L}{2\pi \varepsilon_0 r} \hat{a}_r \ V/m$

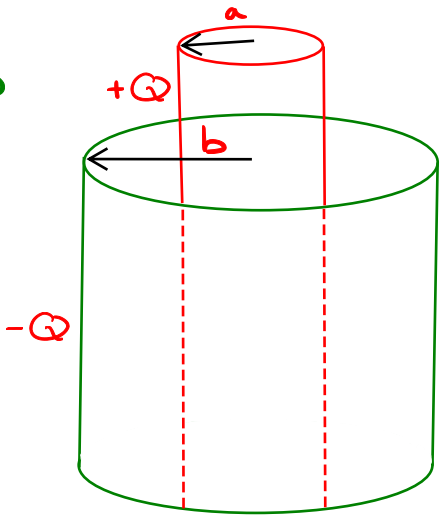
-the variation of D_r with r , for infinite line charge with uniform ρ_L



17). Two concentric cylindrical conductors of radius $a = 0.01\,m$ and $b = 0.08\,m$. The inner cylinder has a charge density $\rho_{sa} = 40\,pC/m^2$ while the outer cylinder has ρ_{sb} such that $\vec{D}$ and $\vec{E}$ field exists between the two cylinders but they are zero elsewhere.

a). Find ρ_{sb}

b). Drive an expression for $\vec{D}$ and $\vec{E}$ between the two cylinders



a). $\vec{D}$ & $\vec{E} = 0$ at $r > b$, $\therefore Q_a = -Q_b$

$\therefore \rho_{sa} 2\pi a L = -\rho_{sb} 2\pi b L$

$\therefore \rho_{sb} = -\rho_{sa}\left(\frac{a}{b}\right) = -40 \times 10^{-12} \times \frac{0.01}{0.08}$

$\rho_{sb} = \boxed{-5\,pC/m^2}$

b). field is in $\hat{a}_r$ direction only

$\Rightarrow \oint \vec{D} \cdot ds = Q_{en}$, $D_r (2\pi r L) = Q_{en}$

for $0 < r < a$: $Q_{en} = 0$, $\vec{D} = 0$, $\vec{E} = 0$

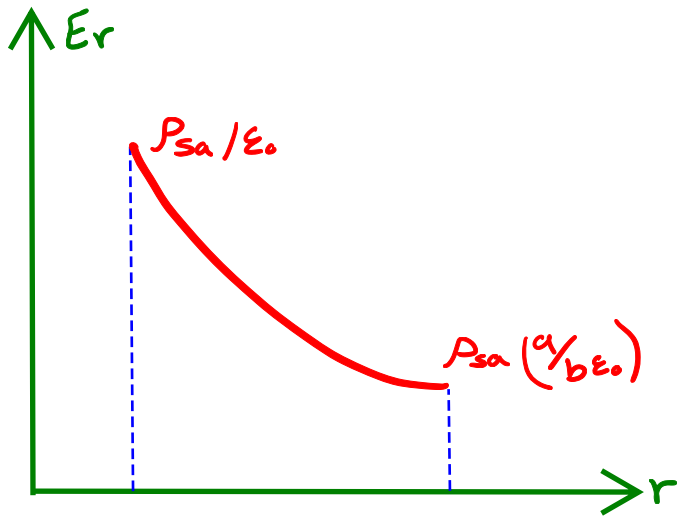
for $a < r < b$: $Q = 2\pi a L (\rho_{sa})$

$\therefore \vec{D}_r = \frac{\rho_{sa} a}{r} = \frac{40 \times 10^{-12} \times 0.01}{r} = \frac{4 \times 10^{-13}}{r}\ C/m^2$

$\therefore \vec{E}_r = \frac{\rho_{sa} a}{\varepsilon_0 r} = \frac{4.52 \times 10^{-2}}{r}\ V/m^2$

for $b<r<\infty$: $Q_{en} = Q - Q = 0$, $\vec{D} = 0$, $\vec{E} = 0$

- the relationship between the field E_r as a function of r



18).Given that D = $\frac{10\,r^3}{4}$ âr C/m² in the region 0 < r < 3 m in cylindrical coordinates and D = $\frac{810}{4r}$ âr C/m² elsewhere. Find the charge density?

∵ $\vec{D}$ is in âr direction only

for $0<r<3$: $\text{Div}\,\vec{D} = \vec{\nabla}\cdot\vec{D} = \frac{1}{r}\frac{\partial(rD_r)}{\partial r} = \frac{1}{r}\frac{\partial}{\partial r}(10r^4/4) =$

$\Rightarrow \rho_v = \frac{40r^3}{4r} = 10r^2 \quad C/m^3$

- for $r>3$: $\text{Div}\,\vec{D} = \vec{\nabla}\cdot\vec{D} = \frac{\partial(810r/4r)}{r\,\partial r} = \frac{1}{r}\frac{\partial}{\partial r}(810/4) = 0$ (constant)

$\Rightarrow \rho_v = 0 \; C/m^3$

19). Given that $\vec{D} = 30e^{-r}\hat{a}_r - 2z\hat{a}_z \ C/m^2$ in Cylindrical Coordinates, evaluate both sides of divergence theorem for the volume enclosed by $r = 2m$, $z = 0$ and $z = 5m$

$\Rightarrow$ divergence theorem: $\oint_S \vec{D}.ds = \int_{Vol} (\nabla.\vec{D})\, dv$

$D_z = 0$ عند $z = 0$، عشان كدة $\vec{D}.d\vec{s}$ هتساوى صفر

first L.H.S: $\oint_S \vec{D}.ds = \int_{side} \vec{D}.ds + \int_{TOP} \vec{D}.d\vec{s}$

$$= \int_0^5 \int_0^{2\pi} D_r \hat{a}_r \ (2)\, d\phi dz \hat{a}_r + \int_0^{2\pi} \int_0^2 D_z \hat{a}_z . r\, dr d\phi \hat{a}_z = \boxed{129.4C}$$

R.H.S: $Div\vec{D} = \frac{1}{r}\frac{\partial (30re^{-r})}{\partial r} + \frac{\partial(-2z)}{\partial z} = \frac{30e^{-r}}{r} - 30e^{-r} - 2$

$$\int_{Vol} (\nabla.\vec{D})\, dv = \int_0^5 \int_0^{2\pi} \int_0^2 \left[\frac{30e^{-r}}{r} - 30e^{-r} - 2\right] r\, dr\, d\phi\, dz = \boxed{129.4C}$$

mid-2020

20). within the spherical shell, $3 < r < 4m$, the electric flux density is given as $\vec{D} = 5(r-3)^3 \hat{a}_r \ C/m^3$. the electric flux density in C/m^2 at $r = 4m$ is?

صعوبة في سؤال:

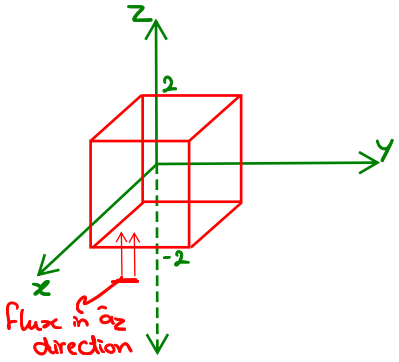
$$\vec{D}|_{r=4} = 5(4-3)^3 \hat{a}_r \frac{C}{m^3}$$

$$= 5\hat{a}_r \frac{C}{m^3}$$

21). An electric field in free space is $\mathbf{E} = (5z^2/\epsilon_0)\,\hat{\mathbf{a}}_z$ V/m. Find the total charge contained within a cube, centered at the origin, of 4-m side length, in which all sides are parallel to coordinate axes (and therefore each side intersects an axis at ±2).

$\vec{D} = \varepsilon_0 \vec{E} = 5z^2 \cdot \hat{a}_z$

$\because \vec{D}$ is in z-direction only , $\because \oint \vec{D} \cdot d\vec{s} = Q_{en}$



note : it will intersect only the top and bottom surfaces

$\therefore Q_{en} = \int_{-2}^{2}\int_{-2}^{2} 5(2)^2 \cdot \hat{a}_z \cdot \hat{a}_z \, dx\,dy + \int_{-2}^{2}\int_{-2}^{2} 5(-2)^2 \cdot \hat{a}_z \cdot (-\hat{a}_z) \, dx\,dy = 0$

22). Volume charge density is located in free space as $\rho_v = 2e^{-1000r}$ nC/m^3 for $0 < r < 1$ mm, and $\rho_v = 0$ elsewhere. (*a*) Find the total charge enclosed by the spherical surface $r = 1$ mm. (*b*) By using Gauss's law, calculate the value of D_r on the surface $r = 1$ mm.

a)

$Q = \iiint \rho_v \, dv = \int_0^{2\pi}\int_0^{\pi}\int_0^{0.001} 2e^{-1000r} r^2 \sin\theta \, dr\,d\theta\,d\phi$

$Q = 2\pi \cdot [-\cos\theta|_0^{\pi}] \left[\frac{r^2 e^{-1000r}}{-1000} \Big|_0^{0.001} + \frac{2}{1000}\frac{e^{-1000r}}{(1000)^2}(-1000r-1)\Big|_0^{0.001} \right]$

$Q = \boxed{4.0 \times 10^{-9}\ \text{nC}}$

for the radial components:

$u = r^2 \quad dv = e^{-1000r}$
$du = 2r \quad v = \frac{e^{-1000r}}{-1000}$

$= \frac{r^2 e^{-1000r}}{-1000} + \frac{2}{1000}\int r\, e^{-1000r}$

$u = r \quad dv = e^{-1000r}$
$du = 1 \quad v = \frac{e^{-1000r}}{-1000}$

$= \frac{r^2 e^{-1000r}}{-1000} + \frac{2}{1000}\left[\frac{r e^{-1000r}}{-1000} - \frac{e^{-1000r}}{(-1000)^2} \right]$

b). Gaussian surface is spherical shell,

$\therefore 4\pi r^2 \vec{D} = Q_{en} \quad , D_r = \frac{Q}{4\pi r^2} = \frac{4.0 \times 10^{-9}}{4\pi (0.001)^2}$

$\therefore D_r = \boxed{3.2 \times 10^{-4}\ \text{nC/m}^2}$

23). a uniform volume Charge density of $\rho_v = 4r^2\ C/m^3$ lies within the spherical $r = 5\ mm$, and ρ_v is Zero else where.

i). Find the electric flux density in all regions.

1). $r < 5\ mm$: $\oint \vec{D} \cdot \vec{ds} = Q_{en} = \int_v \rho_v\, dv$

$$\Rightarrow D \times 4\pi (r)^2 = \int_0^{2\pi}\int_0^{\pi}\int_0^{r} 4r^2 \cdot r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$\therefore D \times 4\pi r^2 = \frac{4}{5} r^5 \cdot (2) \cdot (2\pi)$$

$$\therefore \vec{D}_1 = \boxed{\frac{4}{5} r^3\, \hat{a}_r}$$

2). $r > 5mm$: $D(4\pi r^2) = \int_0^{2\pi}\int_0^{\pi}\int_0^{5\times10^{-3}} 4r^2 \cdot r^2 \sin\theta\, dr\, d\theta\, d\phi$

$$D(4\pi r^2) = (2\pi)(4/5)(2)(5\times10^{-3})^5, \quad \vec{D}_2 = \boxed{\frac{2.5\times10^{-12}}{r^2}\, \hat{a}_r}$$

24). A uniform volume charge density of 80 μC/m^3 is present throughout the region 8 mm < r < 10 mm. Let $\rho_v = 0$ for 0 < r < 8 mm. (*a*) Find the total charge inside the spherical surface r = 10 mm. (*b*) Find D_r at r = 10 mm. (*c*) If there is no charge for r > 10 mm, find D_r at r = 20 mm.

$$Q = \int_0^{2\pi}\int_0^{\pi}\int_{0.008}^{0.010} (80\times10^{-6})\, r^2 \sin\theta\, dr\, d\theta\, d\phi = 4\pi (80\times10^{-6}) \frac{r^3}{3}\Big|_{0.008}^{0.010}$$

$$= \boxed{164\ pC}$$

b). Gaussian Surface is spherical shell: $4\pi r^2 D_r = Q = 164\times10^{-12}$

$$D_r(10mm) = \frac{164\times10^{-12}}{4\pi(0.01)^2} = \boxed{130\ nC/m^2}$$

c). Same as (b), except the gaussian surface now lies at 20 mm

$\therefore D_r(20\,mm) = \frac{164 \times 10^{-12}}{4\pi(0.02)^2} = \boxed{32.5\ nC/m^2}$

25). Within the spherical shell, $3 < r < 4$ m, the electric flux density is given as $\mathbf{D} = 5(r-3)^3\,\mathbf{a}_r$ C/m^2. (*a*) What is the volume charge density at $r = 4$? (*b*) What is the electric flux density at $r = 4$? (*c*) How much electric flux leaves the sphere $r = 4$? (*d*) How much charge is contained within the sphere $r = 4$?

$\vec{D} = 5(r-3)^3\,\hat{a}_r$

a). $\rho_v = \nabla \cdot \vec{D} = \frac{1}{r^2}\frac{d(r^2 D_r)}{dr} = \frac{1}{r^2}\frac{d(5r^2(r-3)^3)}{dr} = \frac{1}{r^2}\left[10r(r-3)^3 + 15r^2(r-3)^2\right]$

$= \frac{5}{r}(r-3)^2(5r-6)\ C/m^3$ $\quad\therefore \rho_v$ at $(r=4) = \boxed{17.5\ C/m^3}$

b). Substitute in $5(r-3)^3\,\hat{a}_r$, $\therefore D(4) = \boxed{5\hat{a}_r\ C/m^2}$

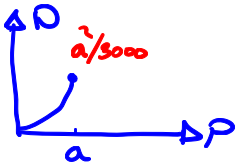
c). using result in Part (b), $\therefore \Psi = 4\pi r^2 \cdot D = 4\pi(4)^2(5) = \boxed{320\pi C}$

d). $\Psi = Q = \boxed{320\pi C}$

26| For the following cylindrical charge distribution $\rho_v = \frac{\rho}{1000}\ \mu C/m^3$ for $0<\rho<a$, $\rho_v = 2\ \mu C/m^3$ for $a<\rho<b$, $\rho_v = 0$ elsewhere. Find the electric flux density and sketch it in all regions.

$0 \le \rho \le a$: $D(2\pi\rho L) = \int \rho_v\, dV = \int_v \frac{\rho}{1000}\, \rho\, d\rho\, d\phi\, dz = \frac{1}{1000}\cdot\frac{\rho^3}{3}\Big|_0^{\rho}(2\pi)(L)$

$\therefore\ D = \frac{\rho^2}{3000}\ \mu C/m^2$

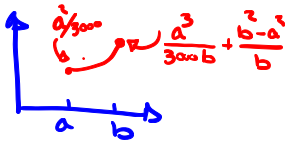


$a \le \rho \le b$:

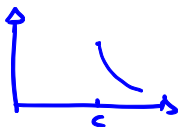
$D(2\pi\rho L) = Q_{0\to b} = \frac{\rho^3}{3000}\Big|_0^a (2\pi)(L) + 2\times\pi(\rho^2 - a^2)L$

Q from $0 \to a$; ρ_v = uniform in $a<\rho<b$ region

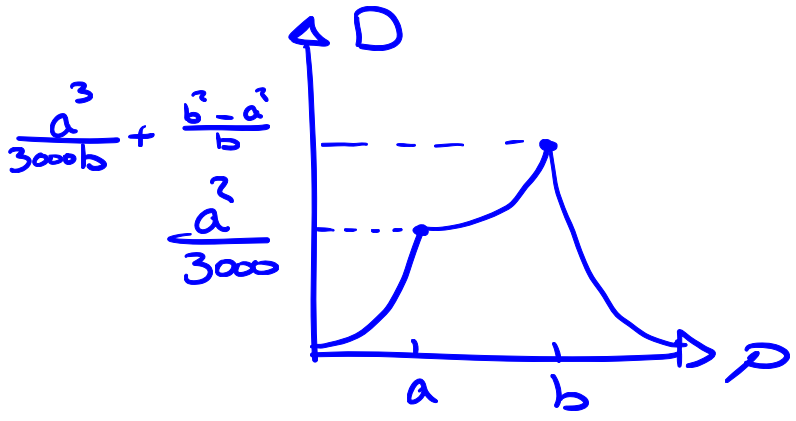
$D = \frac{a^3}{\rho 3000} + \frac{(\rho^2 - a^2)}{\rho}\ \mu C/m^2$ "increasing"



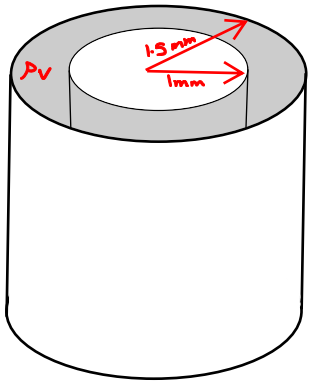
$\rho > b$: $D = \frac{Q}{2\pi\rho L} = \frac{\frac{a^3}{3000} + (b^2 - a^2)}{\rho}$



$\therefore$



27). In cylindrical coordinates, let $\rho_v = 0$ for $\rho < 1$ mm, $\rho_v = 2\sin(2000\pi\rho)$ nC/m^3 for 1 mm $< \rho <$ 1.5 mm, and $\rho_v = 0$ for $\rho > 1.5$ mm. Find **D** everywhere.



$\oint \vec{D}.d\vec{s} = Q_{en}$

for $\rho < 1mm$: $\oint \vec{D}.d\vec{s} = Q_{en} = 0$

$\vec{D} = 0\ a_\rho$

b) for $1mm < \rho < 1.5mm$:

$$\vec{D}(2\pi\rho l) = \iiint \rho_v .dv = \int_0^l \int_0^{2\pi} \int_{0.001}^{\rho} 2\sin(2000\pi\rho)\,\rho\, d\rho\, d\phi\, dz$$

$$\vec{D}(2\pi\rho l) = 2[2\pi][l]\int_{0.001}^{\rho} \sin(2000\pi\rho)\,\rho\, d\rho \times 10^{-9}$$

$u = \rho \quad dv = \sin(2000\pi\rho)$

$du = 1 \quad v = \frac{-\cos(2000\pi\rho)}{(2000\pi)}$

$$= -\rho\frac{\cos(2000\pi\rho)}{(2000\pi)} + \int \frac{\cos(2000\pi\rho)}{(2000\pi)}$$

$$= -\rho\frac{\cos(2000\pi\rho)}{(2000\pi)} + \frac{\sin(2000\pi\rho)}{(2000\pi)^2}$$

$$\vec{D}(2\pi\rho l) = 4\pi l \left[-\rho\frac{\cos(2000\pi\rho)}{(2000\pi)} + \frac{\sin(2000\pi\rho)}{(2000\pi)^2} \right]_{0.001}^{\rho} \times 10^{-9}$$

$$\therefore \vec{D} = \frac{10^{-15}}{2\pi^2\rho}\left[\sin(2000\pi\rho) + 2\pi(1 - 10^3\rho\cos(2000\pi\rho)\right] C/m^2$$

b) for $\rho > 1.5mm$:

$$D(2\pi\rho l) = \int_0^l \int_0^{2\pi} \int_{0.001}^{0.015} \rho_v .dv \qquad \therefore \vec{D} = \frac{2.5\times10^{-15}}{\pi\rho}\ a_\rho\ C/m^2$$

28). Volume charge density is located as follows: $\rho_v = 0$ for $\rho < 1$ mm and for $\rho > 2$ mm, $\rho_v = 4\rho\ \mu C/m^3$ for $1 < \rho < 2$ mm. (*a*) Calculate the total charge in the region $0 < \rho < \rho_1$, $0 < z < L$, where $1 < \rho_1 < 2$ mm. (*b*) Use Gauss's law to determine D_ρ at $\rho = \rho_1$. (*c*) Evaluate D_ρ at $\rho = 0.8$ mm, 1.6 mm, and 2.4 mm.

$$Q = \iiint \rho_v \, dv = \int_0^L \int_0^{2\pi} \int_{0.001}^{0.002} 4\rho \cdot \rho \, d\rho \, d\phi \, dz = 2\pi L \cdot \frac{4\rho^3}{3} \Big|_{0.001}^{0.002}$$

$$Q = \boxed{5.864 \times 10^{-8} L \ \mu C/m}$$

b). D_ρ at $\rho = \rho_1$: $2\pi\rho L \cdot D_\rho = Q$ ← from part (a)

$$Q = \iiint \rho_v \, dv = \int_0^L \int_0^{2\pi} \int_{0.001}^{\rho} 4\rho \cdot \rho \, d\rho \, d\phi \, dz = 2\pi L \cdot \frac{4\rho^3}{3} \Big|_{0.001}^{\rho}$$

$$= \frac{8\pi L}{3} \left[\rho_1^3 - 10^{-9}\right] \mu C$$

$$\therefore D_\rho(\rho_1) = \boxed{\frac{4(\rho_1^3 - 10^{-9})}{3\rho_1} \ \mu C/m^2}$$

- now. D_ρ at $\rho = 0.8$ mm : $Q_{en} = 0$ $\therefore D_\rho(0.8\text{mm}) = \boxed{0}$

- at $\rho = 1.6$ mm : from part (b) :

$$D_{\rho(1.6\text{mm})} = \frac{4\left[(0.0016)^3 - (0.0010)^3\right]}{3[0.0016]} = \boxed{3.6 \times 10^{-6} \mu C/m^2}$$

- at $\rho = 2.4$: from (a) : $Q_{en} = 5.864 \times 10^{-8} \mu C/m$

$$\therefore D_{\rho(2.4\text{ mm})} = \frac{5.864 \times 10^{-8}}{(2\pi(0.024))} = \boxed{3.9 \times 10^{-6} \mu C/m^2}$$

29). Given that $\bar{D} = \frac{5r^2}{4}$ âr C/m² in spherical coordinates, evaluate both sides of the divergence theorem for the volume enclosed by r = 1, and r = 2m.

⇒ the divergence theorem: $\oint \vec{D} \cdot d\vec{s} = \int_{Vol} (\bar{\nabla} \cdot \vec{D})\, dV$

R.H.S:

$$\text{Div}\,\vec{D} = \frac{1}{r^2}\frac{\partial(r^2 D_r)}{\partial r} = \frac{1}{r^2}\frac{\partial(5r^4/4)}{\partial r} = 5r$$

and $$\int_{Vol} (\bar{\nabla} \cdot \vec{D})\, dV = \int_1^2 \int_0^{\pi} \int_0^{2\pi} (5r)\, r^2 \sin\theta \, dr\, d\theta\, d\phi$$

$$= 2\pi \int_1^2 (5r^3)\, dr \int_0^{\pi} \sin\theta \, d\theta = \boxed{75\pi \text{ C}}$$

⇒ L.H.S:

$$\oint_S \vec{D} \cdot d\vec{s} = \int_{outer} \vec{D} \cdot d\vec{s} + \int_{inner} \vec{D} \cdot d\vec{s}$$

$$= \int_0^{\pi}\int_0^{2\pi} D_r \hat{a}_r \cdot \underbrace{r^2}_{\to r=2} \sin\theta\, d\theta\, d\phi\, \hat{a}_r + \int_0^{\pi}\int_0^{2\pi} D_r \hat{a}_r \cdot (-\underbrace{r^2}_{\to r=1} \sin\theta\, d\theta\, d\phi)\hat{a}_r$$

$$= \int_0^{\pi}\int_0^{2\pi} \left[\frac{5}{4}\right](2)^2 \sin\theta\, d\theta\, d\phi - \int_0^{\pi}\int_0^{2\pi} \left[\frac{5}{4}\right](1)^2 \sin\theta\, d\theta\, d\phi$$

$$= 20\,[2][2\pi] - [5/4][2][2\pi] = \boxed{75\pi C}$$

30). A spherical surface with a uniform volume charge density of 80 μC/m³ is present throughout the region 8 mm < r < 10 mm. Let $\rho_v = 0$ elsewhere.

(a) The value of D_r at r = 4 mm is.

i. 0 c/m² ii. 10 c/m². iii. -10 c/m² iv. 20 c/m²

(b) The value of D_r at r = 9 mm is.

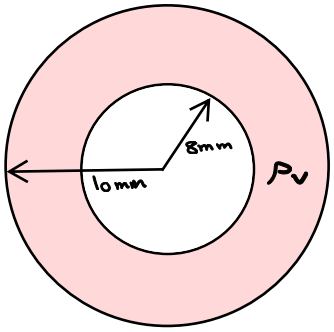
i. 20.4 nc/m² ii. 71.4 nc/m² iii. 100.4 nc/m² iv. 50.4 nc/m²

(c) The value of D_r at r = 20 mm is.

I. 32.5 nc/m² ii. 44.5 nc/m² iii. 71.4 nc/m² iv. 50.4 nc/m²

a). r = 4 mm:

$$\oint \vec{D}.d\vec{s} = Q_{en} = 0 \quad , \quad \therefore \boxed{D_r = 0}$$



b). r = 9 mm:

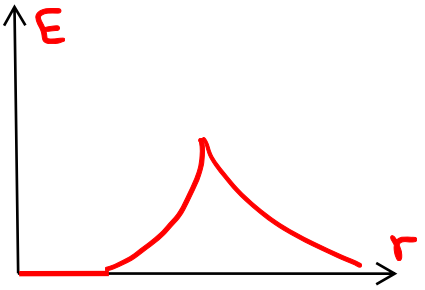
$$4\pi r^2 . D_r = \rho_v \left[\frac{4}{3}\pi r^3 - \frac{4}{3}\pi\, 0.008^3 \right]$$

$$\therefore D_r = \frac{80\times10^{-6}\left[r^3 - 0.008^3\right]}{3r^2} \quad , \quad \therefore D_r(r=9mm) = \boxed{71.4\ nC/m^2}$$

c). r = 20 mm:

$$4\pi r^2 . D_r = \rho_v \left[\frac{4}{3}\pi (0.01)^3 - \frac{4}{3}\pi (0.008)^3 \right]$$

$$\therefore D_r = \frac{1.3\times10^{-11}}{r^2} \quad , \quad \therefore D_r = \boxed{32.5\ nC/m^2}$$



31). An infinitely long cylindrical dielectric of radius b contains charge within it's volume of density ρ_v = 80 μC/m³. Find the electric field strength, E, both inside and outside the cylinder.

⇒ first if ρ < b: $\oint \vec{D}.d\vec{s} = Q_{en}$

$$D(2\pi\rho l) = \rho_v . V \quad , \quad D(2\pi\rho l) = \rho_v . [\pi\rho^2 l]$$

$$\therefore \vec{D} = \frac{\rho_v \rho}{2}\hat{a}_\rho \quad , \quad \therefore \vec{E} = \frac{\vec{D}}{\varepsilon_r\varepsilon_0} = \frac{4\times10^{-5}(\rho)}{8.854\times10^{-12}\varepsilon_r} = \boxed{\frac{4.52\times10^6 \rho\, \hat{a}_\rho}{\varepsilon_r}}$$

⇒ for ρ > b:

$$D(2\pi\rho l) = \rho_v . \pi b^2 l \quad , \quad D = \frac{\rho_v . b^2}{2\rho}$$

$$\therefore \vec{E} = \frac{\vec{D}}{\varepsilon_r\varepsilon_0} = \boxed{\frac{4.52\times10^6 (b^2)}{\rho\varepsilon_r}\hat{a}_\rho}$$

32). For the vector field, $E = 10e^{-\rho}\,\hat{a}\rho - 3z\,\hat{a}z$, verify divergence theorem for the cylindrical region enclosed by r = 2, z = 0 and z = 4.

$\vec{D} = \varepsilon_0 \vec{E} = 10\varepsilon_0 e^{-\rho} a_\rho - 3\varepsilon_0 z\, a_z$

$\Rightarrow$ L.H.S :

$$\oint \vec{D}.d\vec{s} = \int_0^4\int_0^{2\pi} 10\varepsilon_0 e^{-\rho}\,\rho\, d\phi\, dz + \int_0^{2\pi}\int_0^2 -3\varepsilon_0 z\,\rho\, d\rho\, d\phi$$

($\rho = 2$; $z = 4-0 = 4$)

$$= 10\varepsilon_0[2e^{-2}][2\pi][4] - 3\varepsilon_0 4 \cdot \frac{\rho^2}{2}\Big|_0^2 \cdot [2\pi]$$

($\frac{2^2}{2} = 2$)

$\therefore$ L.H.S $= \varepsilon[160\pi e^{-2} - 48\pi] =$ **$-82.77\,\varepsilon$**

$\Rightarrow$ R.H.S :

$$\text{Div}\,\vec{D} = \nabla.\vec{D} = \frac{1}{\rho}\frac{\partial(\rho D_\rho)}{\partial\rho} + \frac{\partial D_z}{\partial z}$$

$$= \frac{1}{\rho}\frac{\partial(\rho e^{-\rho})}{\partial\rho} + \frac{\partial[3z]}{\partial z} = \frac{10\varepsilon_0}{\rho}[e^{-\rho} - \rho e^{-\rho}] - 3\varepsilon_0$$

$$\therefore \iiint(\nabla.\vec{D}).dV = \int_0^4\int_0^{2\pi}\int_0^2 \left[\frac{10\varepsilon_0}{\rho}[e^{-\rho} - \rho e^{-\rho}] - 3\varepsilon_0\right]\rho\, d\rho\, d\phi\, dz$$

$$= [10\varepsilon_0 . \rho e^{-\rho} - 3\rho\varepsilon_0\,|_0^2] . [2\pi] . [4-0]$$

$\therefore$ R.H.S = **$-82.77\,\varepsilon_0$**

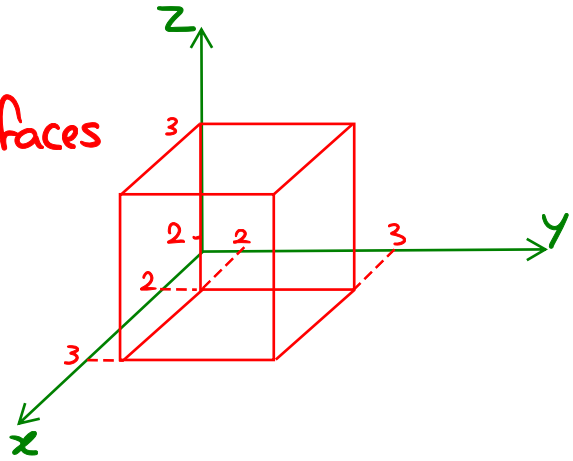
$\therefore$ R.H.S = L.H.S

33). In the region of free space that includes the volume $2 < x, y, z < 3$, $\mathbf{D} = \frac{2}{z^2}(yz\,\mathbf{a}_x + xz\,\mathbf{a}_y - 2xy\,\mathbf{a}_z)$ C/m^2. (a) Evaluate the volume integral side of the divergence theorem for the volume defined here. (b) Evaluate the surface integral side for the corresponding closed surface.

a). $\nabla \cdot \vec{D} = \frac{8xy}{z^3}$, $\iiint \nabla \cdot \vec{D}\, dv = \int_2^3\int_2^3\int_2^3 \frac{8xy}{z^3}\, dx\,dy\,dz$

$= 8\left.\frac{x^2}{2}\right|_2^3 \cdot \left.\frac{y^2}{2}\right|_2^3 \cdot \left.\frac{z^{-2}}{2}\right|_2^3 = \boxed{3.47\,C}$

b) $\oint \vec{D}\cdot d\vec{s}$ = integrate over all six surfaces



$net\Psi = 0$ ∴ x مع متغيرة مش $\hat{a}_x$ ∵ *

$net\,\Psi = 0$ ∴ y مع متغيرة مش $\hat{a}_y$ ∵ *

$\oint \vec{D}\cdot d\vec{s} = \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{left}} + \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{right}} + \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{Front}} + \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{back}} + \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{top}} + \underbrace{\oint \vec{D}\cdot d\vec{s}}_{\text{bottom}}$

Cancel each other — Cancel each other

∴ $\oint \vec{D}\cdot d\vec{s} = \int_2^3\int_2^3 \frac{-4xy}{3^2}\, dx\,dy - \int_2^3\int_2^3 \frac{-4xy}{2^2}\, dx\,dy$

$= (9-4)(9-4)\left[\frac{1}{4} - \frac{1}{9}\right] = \boxed{3.47\,C}$

34). Given the flux density $\mathbf{D} = \frac{16}{r}\cos(2\theta)\,\mathbf{a}_\theta$ C/m^2, use two different methods to find the total charge within the region $1 < r < 2$ m, $1 < \theta < 2$ rad, $1 < \phi < 2$ rad.

$$Q_{en} = \oint \vec{D}\cdot d\vec{s}$$

$$\therefore \oint \vec{D}\cdot d\vec{s} = \int_1^2\int_1^2 \frac{16}{r}\cos(2\theta)\,a_\theta \cdot r\sin\theta\, dr\, d\phi\, a_\theta$$

$$= \int_1^2\int_1^2 \frac{16}{r}\cos(2\theta)\cdot r\sin\theta\, dr\, d\phi \quad (1<\theta<2)$$

(constants)

$$= 16\cos(2\theta)\sin(\theta)\cdot r\Big|_1^2 \cdot \phi\Big|_1^2$$

$$Q = 16\cos(2(2))\sin(2) - 16\cos(2)\sin(1) = \boxed{-3.907\ C}$$

$\Rightarrow$ Alternative methode:

$$\rho_v = \nabla\cdot\vec{D} = \frac{1}{r\sin\theta}\frac{d}{d\theta}(\sin\theta\, D_\theta)$$

$$= \frac{1}{r\sin\theta}\frac{d}{d\theta}\left(\frac{16}{r}\cos(2\theta)\sin(\theta)\right) = \frac{16}{r^2\sin\theta}\left[\cos(2\theta)\cos(\theta) - 2\sin(\theta)\sin(2\theta)\right]$$

$$= \frac{16}{r^2}\left[\frac{\cos(2\theta)\cos(\theta)}{\sin(\theta)} - 2\sin(2\theta)\right]$$

$$Q = \iiint \rho_v\, dv = \iiint \rho_v \cdot r^2\sin\theta\, dr\, d\theta\, d\phi$$

$$Q = \iiint 16\left[\cos(2\theta)\cos(\theta) - 2\sin(\theta)\sin(2\theta)\right] dr\, d\theta\, d\phi$$

$$Q = 16\cdot r\Big|_1^2 \cdot \phi\Big|_1^2 \cdot \int \frac{1}{2}\left[3\cos 3\theta - \cos\theta\right] d\theta = \boxed{-3.907\ C}$$